

Project: Aegis-Grid

4-Bit Binary Target Tracker

Group Members

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The Objective

To build a hardware system that tracks a target (like a laser or finger) across a 16-position grid (4x4) and only **"Fires"** an alarm when the target hits a specific coordinate pre-set using switches.

1. The Input: How We "See" the Target

Instead of 16 sensors, we use a Matrix of 8 sensors (4 for the X-axis/Columns and 4 for the Y-axis/Rows).

- **Sensors:** LDRs (Light Dependent Resistors) or IR pairs.
- **The Math:** 4 sensors on X & 4 sensors on Y. A unique 4x4 coordinate for any spot on the grid.

2. The Processing: How We "Calculate" Coordinates

Two 74148 Priority Encoders turn those 8 sensor wires into binary.

- **Encoder 1 (X):** Turns the 4 column wires into 2 bits.
- **Encoder 2 (Y):** Turns the 4 row wires into 2 bits.
- **The Result:** These are stitched together to get a **4-bit Binary Coordinate (B)** (e.g., Row 2, Col 3 becomes 1011).

3. The Brain: Magnitude Comparison

A 7485 Magnitude Comparator compares the target's real-time position (B) against our "Guard Position" (A).

- **Guard Position (A):** Set using a 4-DIP Switch.
- **If A > B:** Yellow LED lights up (Target is to the Left/Up).
- **If A < B:** Blue LED lights up (Target is to the Right/Down).
- **If A = B:** Green LED lights up (Target Locked).

4. The Output: The "Fire" Protocol

A 7408 AND Gate acts as a safety trigger for the final output.

- **Input 1:** The "Locked" (A=B) signal from the comparator.
- **Input 2:** A "Master Arm" toggle switch.
- **Action:** The system only "Fires" (Buzzer/Red LED) if the target is in the exact right spot AND the safety switch is flipped ON.

Component Summary & Implementation

The table below lists all components used in the Aegis-Grid system along with their specific role and implementation details.

Component	Qty	Role in Project	Implementation Details
74148 IC (Priority Encoder)	2	Priority Encoders (X and Y axes)	Each IC reads 4 active-low sensor inputs and encodes the highest-priority active line into a 2-bit binary output. EI pin is grounded to enable. Encoder 1 handles columns (X), Encoder 2 handles rows (Y). Outputs A0/A1 are combined to form a 4-bit coordinate.
7485 IC (Magnitude Comparator)	1	4-bit Magnitude Comparator	Compares 4-bit input B (live position from encoders) against 4-bit input A (guard position from DIP switch). Cascading inputs set to 0,0,1 for standalone use. Three outputs drive Yellow, Blue, and Green LEDs.
7408 IC (AND Gate)	1	AND Gate (Fire/Safety Logic)	One gate used. Pin 1 receives A=B (Locked) output from 7485. Pin 2 connected to Master Arm toggle switch (HIGH when armed). Output on Pin 3 goes HIGH only when both conditions are true, triggering buzzer and Red LED.
7404 IC (NOT Gate)	1	NOT Gate (Logic Inverter)	Used to invert the active-low outputs of the 74148 encoders into active-high signals compatible with the 7485 comparator inputs.
LDRs (Light Sensors)	8	Grid Sensors (Position Detection)	Arranged in a 4x4 grid — 4 LDRs for column detection (X-axis) and 4 for row detection (Y-axis). Each paired with a 10kΩ pull-down resistor. When light hits an LDR, resistance drops, pulling sensor node HIGH.
4-DIP Switch	1	Manual Target Entry (Guard Position A)	Each switch represents one bit of the 4-bit guard position (A3-A0). When ON, connects to Vcc (5V) through a pull-up resistor, setting that bit HIGH. Wired to A inputs of 7485 comparator.
Standard Breadboard	1	Main Circuit Base	All ICs, sensors, resistors, and LEDs mounted here. Power rails carry regulated 5V from 7805 regulator. ICs placed to minimize long wire runs.
LEDs (Assorted)	6	Status Indicators (Left, Right, Locked, Fire)	Yellow: A>B output via 330Ω. Blue: A<B output. Green: A=B output. Red: AND gate output (Fire signal). Each LED has cathode to GND and anode to signal through 330Ω resistor.
Resistor Pack	1	Pull-down & LED Current Limiters	10kΩ resistors in voltage dividers for each LDR. 330Ω resistors limit current through each LED to ~10mA at 5V.

Component	Qty	Role in Project	Implementation Details
Connecting Wires	1m	Internal Circuit Wiring	Jumper wires connect ICs, sensors, switches, and output components. Color coding recommended: red for Vcc, black for GND.
9V Battery + Clip	1	Power Source	Provides raw input voltage. Feeds into 7805 regulator. Decoupling capacitor (0.1μF) placed close to regulator to stabilize input voltage.
7805 Regulator	1	Voltage Regulator (9V to 5V)	Steps down 9V battery to stable 5V for all TTL ICs. Input pin to 9V, output pin to positive power rail, ground pin to common GND. Output capacitor (0.1μF) ensures stability.